Statement of Purpose

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B.Tech, Computer Science & Engineering

Though many areas in artificial intelligence interest me, for my doctoral studies, I am interested in gaining a thorough understanding of natural language processing and artificial intelligence research questions at the intersection of machine learning and public policy. After my doctoral studies, I intend to pursue a research career in academia or industry.

My first brush with research happened during my sophomore year at my undergraduate program (February, 2021) when I started working on a project focused on developing algorithms and deploying and distributing Deep Learning Models and layers on low computing edge devices with Dr. Amitava Mukherjee (Adjunct Professor, SUNY Polytechnic Institute, College of Nano Science and Engineering, Albany, NY) and Dr. Arnab Raha (Staff Research Scientist, Advanced Architecture Research, Intel AI). In this project, we proposed a greedy, graph based solution different from existing existing methods with comparable (in a few cases 3-5% better) performance. Working first hand on this project and witnessing different stages of research right from the very beginning of figuring out the problem statement to investigating viable solution paths and exploring related literature in parallel, inspired me to broaden my horizon by trying adjacent machine learning sub-fields .

While continuing my project under the mentorship of Prof. Mukherjee and Dr. Raha, I was fortunate to work on a natural language processing (NLP) project under the mentorship of Dr. Asif Ekbal (Department of CSE, Indian Institute of Technology Patna). In this project, I worked on enhancing persona and generating perception-aware dialogues with NLP procedures and infusing knowledge using Knowledge-Graphs. I learnt many advanced and new concepts like Knowledge Graph, Graph Transformer and Graph Neural Networks besides my existing knowledge in Recurrent Neural Networks and Transformers, and I became truly fascinated with NLP methods which led to my next NLP project with Prof. Utpal Garain from Indian Statistical Institute, Kolkata on hate speech recognition and hate phrase extraction in my native language of Bengali, a low-resource language, using the HASOC dataset.

After gaining substantial experience through my two NLP projects, I reached out to Prof. KhudaBukhsh (Rochester Institute of Technology) for mentorship. I grew a keen interest

in the applied side of NLP. Reading the paper “We Don't Speak the Same Language: Interpreting Polarization through Machine Translation” by Dr. Ashiqur Khudabukhsh had a transformative effect in me. His unique and innovative research perspective and methodology combined with a general theme of AI for Social Good is inspiring. Intrigued by his work, I wrote to him and fortunately got the chance to work with him on a project that introduced the novel concept of web archeology. Dr. KhudaBukhsh gave me access to a unique dataset consisting of comments on YouTube channels deleted after the Capitol riot. We employed a broad range of cutting-edge NLP methods to glean insights from these deleted comments and observed disturbing discourse around election fraud, vaccine misinformation, and call for violence. We are on track to submitting this work (jointly done with Adel Khorramrouz and Sujan Dutta, MS and Ph.D. students of Dr. KhudaBukhsh, respectively) to SIGIR 2023 (Perspective Track) or IJCAI 2023 (AI for Good Track).

I informally audited Dr. KhudaBukhsh and Prof. Mark Kamlet’s RIT-CMU course on machine learning on political data. This seminar-style course gave me an exciting opportunity to familiarize myself with seminal papers at the intersection of machine learning, political science, and natural language processing. It was also a thrilling experience to see my project partner Adel Khorramrouz present our preliminary findings to Prof. Kamlet, a noted political scientist and economist during the final class project presentation.

A doctoral program in Computing and Information Sciences at the Rochester Institute of Technology under the supervision of Dr. KhudaBukhsh is a great opportunity for furthering my research career for the following reasons. First, I have already worked under his mentorship and I feel that our research interests align perfectly. Second, I have already collaborated with his students. Dr. KhudaBukhsh’s group consists of highly talented researchers, who are extremely enthusiastic about AI for Good problems and are willing to help each other, making it a great home for early career researchers. Third, I am also aware of Dr. KhudaBukhsh’s ongoing collaborations with Prof. Christopher Homan. I feel that Prof. Homan and Prof. KhudaBukhsh’s research programs have a lot of synergy. I would be thrilled to work on projects co-investigated by them. Finally, RIT being a great champion of diversity and inclusion, and a top computer science place, I am sure I will have a thriving student community to learn from and learn together.

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